

CCE1 EXAMINATION QUESTION BANK — MASTER CONSOLIDATED SOLUTIONS

Course Pattern: TE IT - 2024 Pattern (SPPU)**Academic Term:** Semester V (2025-26 / 2026-27)**Document Type:** Master Consolidated Solutions (Reundancy Grouped)
Course Code: OE329COMT / OE-329**IMPORTANT:**

Consolidated Question Bank Architecture: This master document groups and unifies all overlapping and redundant questions from the official CCE1 Question Bank (covering all 20 questions across Unit 1 and Unit 2) into **8 Comprehensive Topic Solutions**. All mapped question numbers, BTL levels, and marks weightage are clearly delineated.

Unit 1: Introduction to Intellectual Property Rights (CO302.1)

Topic 1.1: IPR Meaning, Concept, Major Types & Strategic Importance in Engineering

GROUP:**Consolidated Questions Covered:**

- **Q1:** Explain meaning, concept, types, and importance of IPR in engineering and innovation. [BTL2, 5 Marks]
- **Q3:** Explain need for IPR in protecting engineering designs with examples from electronics, electrical, software. [BTL2, 5 Marks]
- **Q5:** Explain major types of IPR and distinguish relevance to engineering innovations. [BTL2, 5 Marks]
- **Q6:** Explain role of IPR in encouraging research, innovation, technology transfer, commercialization, investment. [BTL2, 5 Marks]

1. Meaning & Statutory Concept of IPR: Intellectual Property Rights (IPR) are legal and statutory monopoly rights conferred by a sovereign state upon creators, inventors, and enterprises over creations of human intellect (inventions, software designs, literary/artistic expressions, distinctive signs, and trade secrets). These rights grant exclusive legal control to commercialize the intellectual asset for a predetermined statutory duration while legally prohibiting unauthorized exploitation by competitors.

2. Comprehensive Parameter-Based Matrix of Major IPR Types:

IPR Type	Subject Matter Protected	Statutory Duration	Engineering Innovation Example
Patent	Novel, non-obvious technological inventions and functional processes	20 Years from filing date	Novel lithium-solid-state battery electrolyte chemistry
Copyright	Original literary expression, source code, firmware, schematics	Lifetime + 60 Years (Individual) / 60 Years (Corp)	Operating system kernel source code, embedded C firmware

IPR Type	Subject Matter Protected	Statutory Duration	Engineering Innovation Example
Trademark	Distinctive signs, brand names, logos, domain marks	10 Years (Renewable indefinitely)	"NVIDIA", "Snapdragon", "TensorFlow" brand marks
Industrial Design	Aesthetic non-functional 2D/3D shapes, patterns, configurations	10 Years + 5 Years extension (15 Years Max)	Ergonomic outer chassis contours of Apple MacBook
Trade Secret	Confidential formulas, manufacturing methods, proprietary datasets	Indefinite (as long as kept strictly secret)	Google Search Ranking algorithm, proprietary catalyst formula

3. Need & Domain-Specific Applications:

- **Electronics & VLSI:** Semiconductor Integrated Circuits Layout-Design Act (SICLDA), 2000 & Hardware Patents protect ASIC architectures against chip decapsulation cloning.
- **Electrical Systems:** Utility patents protect novel regenerative motor controllers; industrial designs protect transformer enclosures.
- **Software Engineering:** Copyright Act, 1957 protects literal source code; Patents protect technical hardware-interfacing algorithms; Trade Secrets protect deep learning weights.

4. Role in R&D, Technology Transfer & Commercialization:

- **Incentivizing R&D:** 20-year exclusivity ensures recovery of massive R&D capital expenditure.
- **Technology Transfer:** Standard Essential Patents (SEPs) enable global technology standards (Wi-Fi, 5G) under FRAND licensing.
- **Venture Capital Attractiveness:** Defensible patent portfolios serve as essential collateral for startup investments.

Source: Deborah Bouchoux, Ch 1, p. 1-30 | Source: Dr. M.K. Bhandari, Ch 1, p. 1-28

Topic 1.2: International IPR Treaties & Frameworks (WIPO, WTO & TRIPS Impact on India)

GROUP:

Consolidated Questions Covered:

- **Q2:** Apply roles of WIPO and WTO to explain relevance to IPR in engineering design in India. [BTL3, 5 Marks]
- **Q8:** Explain functions of WIPO and its contribution to international protection of IP. [BTL2, 5 Marks]
- **Q9:** Apply WTO-TRIPS framework to explain how international standards influence IPR in India. [BTL3, 5 Marks]

1. WIPO (World Intellectual Property Organization) Functions:

- Specialized UN agency administering 26 international treaties (Paris Convention, Berne Convention, PCT).
- **Patent Cooperation Treaty (PCT):** Allows Indian engineers to file one unified international patent application, securing priority dates across 155+ member nations.
- **WIPO ADR Center:** Neutral international arbitration for cross-border technology licensing disputes.

2. WTO & TRIPS Agreement Foundations:

- **National Treatment (Art 3):** Equal treatment for foreign and domestic IP holders.
- **Most-Favoured-Nation (MFN - Art 4):** Privileges granted to one nation must be extended to all WTO members.

3. Direct Influence on Indian IPR Legislation:

- **Product Patents:** Patents (Amendment) Act, 2005 introduced full product patents for pharmaceuticals and chemical substances (repealing Section 5).
- **Uniform 20-Year Term:** Standardized all patent terms to 20 years from filing.

- **Section 3(d) Safeguards:** Prevents evergreening of patents by rejecting minor modifications lacking enhanced therapeutic/technical efficacy.

Source: Dr. M.K. Bhandari, Ch 2, p. 45-82

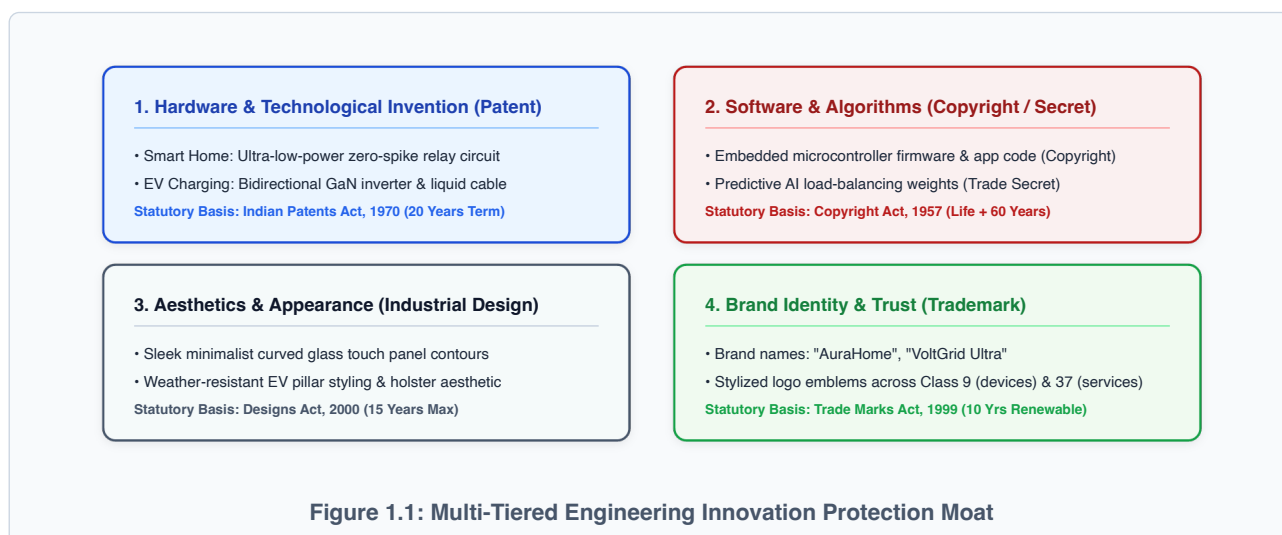
Topic 1.3: Scenario-Based Applied IPR Protection (Smart Home, EV Charging & Innovations)

GROUP:

Consolidated Questions Covered:

- **Q4:** Apply IPR concepts to an engineering innovation and identify forms of protection. [BTL3, 5 Marks]
- **Q7:** Apply IPR concepts to a newly developed smart home automation system (hardware, software, design, brand). [BTL3, 5 Marks]
- **Q10:** Apply IPR concepts to an innovative electric vehicle charging system. [BTL3, 5 Marks]

1. Integrated Multi-Layered Protection Framework:



Source: Stim, Ch 1 & 10, p. 5-35

Unit 2: Patents, Copyrights, Trademarks & Industrial Designs (CO302.2)

Topic 2.1: Patent Foundations, Criteria, Grant Procedure in India & Patentee Rights

GROUP:

Consolidated Questions Covered:

- **Q1:** Explain concept, types, patentability criteria, rights, and Indian registration procedure with examples. [BTL2, 5 Marks]
- **Q5:** Explain concept and types of patents and major patentability criteria with engineering examples. [BTL2, 5 Marks]
- **Q6:** Explain major steps in obtaining a patent in India from filing to grant and state patentee rights. [BTL2, 5 Marks]

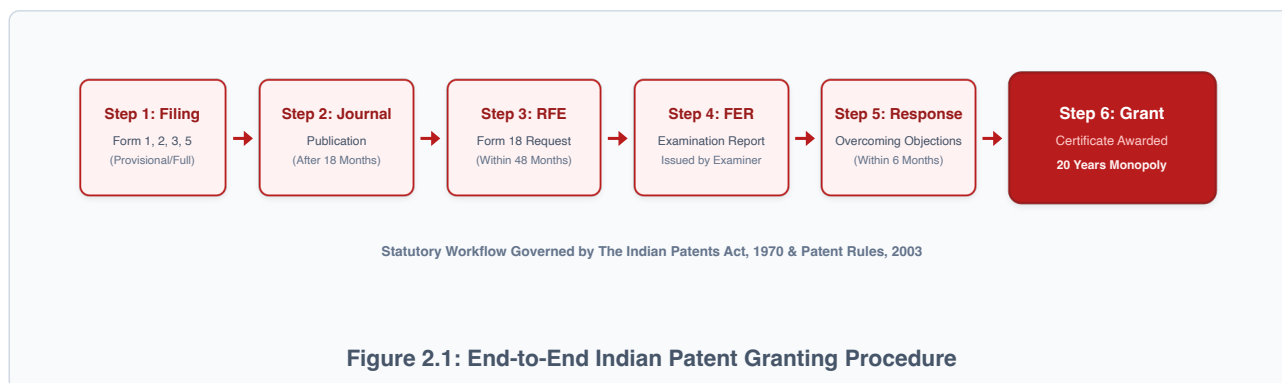
1. Statutory Concept & Types of Applications: A Patent is an exclusive 20-year legal monopoly granted under the Indian Patents Act, 1970 for novel, non-obvious, and industrially applicable inventions in exchange for full public disclosure.

- **Types:** Ordinary Application, Convention Application (Paris Convention priority), PCT National Phase Application, Patent of Addition (for improvements).

2. Three Statutory Patentability Criteria (Section 2(1)(j)):

1. **Novelty:** Must not be anticipated by prior publication or use anywhere in the world (Prior Art).
2. **Inventive Step (Non-Obviousness):** Technical advance non-obvious to a person skilled in the relevant art.
3. **Industrial Applicability:** Must be capable of industrial manufacture and physical repeatability.

3. End-to-End Patent Registration Workflow in India:



4. Rights of Patent Holder (Section 48):

- Exclusive right to prevent unauthorized making, using, selling, or importing the patented product/process.
- Statutory right to assign, transfer, or license the patent for royalties.

Source: Dr. M.K. Bhandari, Ch 4, p. 85-132

Topic 2.2: Comparative Analysis of IPR Disciplines (Patents, Copyrights, Trademarks, Designs, GIs)

GROUP:

Consolidated Questions Covered:

- **Q2:** Apply concepts of copyright, trademark, industrial design, GI to engineering examples and protection provided. [BTL3, 5 Marks]
- **Q10:** Explain industrial design and GI; distinguish them from patents and trademarks with examples. [BTL2, 5 Marks]

Parameter-Based Comparison Across All Major IPR Disciplines:

Parameter	Industrial Design	Geographical Indication	Patent	Trademark
Primary Protection	Aesthetic visual exterior shape & styling	Regional origin & traditional product quality	Functional technical invention & process	Commercial brand identity, logo, and source
Functionality Covered?	Strictly NO (Purely non-functional aesthetics)	NO (Regional pedigree & reputation)	YES (Core functional mechanism)	NO (Brand differentiation only)

Parameter	Industrial Design	Geographical Indication	Patent	Trademark
Ownership	Single Enterprise / Individual Designer	Community of producers in designated region	Inventor / Assignee Enterprise	Individual, Company, or Trust
Statutory Term	15 Years Maximum (10 + 5)	10 Years (Renewable indefinitely)	20 Years Fixed from Filing Date	10 Years (Renewable indefinitely)
Representative Example	Coca-Cola contour glass bottle shape	Darjeeling Tea, Banarasi Silk	Novel fuel-injection pump mechanism	"Coca-Cola" Spencerian script logo

Source: Dr. M.K. Bhandari, Ch 6 & 7, p. 185-220

Topic 2.3: Software & Embedded Systems Copyright Protection & Trademark Registration

GROUP:

Consolidated Questions Covered:

- **Q3:** Scope of copyright for software and embedded systems; describe trademark registration with branding examples. [BTL2, 5 Marks]
- **Q8:** Explain copyright protection for software and embedded systems; illustrate scope of protection with examples. [BTL2, 5 Marks]

1. Scope of Software Copyright (Indian Copyright Act, 1957 - Section 2(o)):

- **Protected Scope:** Literal source code, compiled binaries, embedded microcode on microcontrollers, user manual documentation, and UI visual iconography (as artistic works).
- **Idea-Expression Dichotomy:** Algorithms, mathematical logic, and system architectures are NOT protected by copyright; only the literal written code expression is protected against verbatim copying and piracy.
- **Duration:** Lifetime of author + 60 years.

2. Trademark Registration & Protection (Trade Marks Act, 1999):

- Distinctive signs, emblems, phrases, or sound marks identifying commercial origin.
- **Procedure:** Trademark search across 45 classes → Form TM-A filing → Examination → Journal publication for 4-month opposition → 10-year renewable registration.
- **Examples:** Intel inside logo, Android green robot logo, Microsoft Windows wordmark.

Source: Deborah Bouchoux, Ch 6, p. 110-135

Topic 2.4: Patentability Evaluations for Engineering Innovations (Solar Irrigation & Drone)

GROUP:

Consolidated Questions Covered:

- **Q4:** Apply patentability criteria to an engineering innovation and identify multi-disciplinary protection. [BTL3, 5 Marks]
- **Q7:** Apply patentability criteria to an innovative solar-powered irrigation system. [BTL3, 5 Marks]

1. Detailed Case Evaluation: Solar-Powered Irrigation System:

- **Novelty Test:** Closed-loop MPPT pump modulation coupled dynamically with soil salinity/moisture feedback has not been anticipated in prior art. (**Passed**)

- **Inventive Step Test:** Modulating solar charge parameters directly from soil impedance curves represents a non-obvious engineering advancement that saves 40% battery capacity. **(Passed)**
- **Industrial Applicability:** Repeatable manufacturing and deployment across farms worldwide. **(Passed)**
- **Section 3(k) Check:** Produces a tangible technical effect on physical hardware (modulating pump and water valves), overcoming pure software exclusions.
- **Verdict: Fully Qualifies for Patent Protection.**

Source: Deborah Bouchoux, Ch 2, p. 40-62

Topic 2.5: Comprehensive Multi-Moat IPR Strategy for Smartphone Products

GROUP:

Consolidated Questions Covered:

- **Q9:** Apply appropriate IPR protection to a smartphone product (invention, software, appearance, brand name) with justifications. [BTL3, 5 Marks]

1. Smartphone Multi-Tiered Moat Strategy:

Smartphone Asset	Specific Component Details	IPR Mechanism	Strategic Justification
Technical Inven-tions	Under-display optical fingerprint scanner; 120W fast-charging circuit; 5G beamforming an-tenna	Patents (Patents Act, 1970)	Prevents rival smartphone manufacturers from copy-ing high-tech internal circuits and physical sensor innova-tions.
Software & Firmware	Mobile Operating System kernel code, camera image signal pro-cessing firmware, default apps	Copyright (Copyright Act, 1957)	Provides immediate global protection against unautho-rized copying, reverse-engi-neering, and pirated distrib-ution of OS code.
External Appear-ance	Curved bezel-less glass display curvature, distinct rear camera island layout, button chamfers	Industrial Designs (De-signs Act, 2000)	Prevents competitors from manufacturing visually iden-tical cosmetic look-alike phone shells in the market.
Brand & Commer-cials	Brand name "Galaxy / iPhone", company logo, signature boot chime sound mark	Trademarks (Trade Marks Act, 1999)	Secures brand identity, con-sumer goodwill, and pre-vents counterfeit devices from deceiving buyers.

Source: Stim, Ch 10, p. 200-225